Test-ExecutableSafety

Combined safety check for a downloaded Windows program

Source: Test-ExecutableSafety.md · Generated 2026-11-12

How to run the script

1. Save `Test-ExecutableSafety.ps1` to a folder on your PC, e.g. `Documents\Checks`.
2. Open the Start menu, type **PowerShell** and open **Windows PowerShell**.
3. Type the command below, with the path to the downloaded file you want to check, and press Enter:

```
powershell -ExecutionPolicy Bypass -File "$env:USERPROFILE\Documents\Checks\Test-
ExecutableSafety.ps1" -Path "$env:USERPROFILE\Downloads\setup.exe"
```

The script only reads the file; it never runs it. The sections below explain every result and what to do about it.

A PowerShell script that checks a downloaded Windows executable **before you run it**. It never executes the file. It reads the file, its signature, its metadata and the machine's security settings, then gives a verdict.

Split versions with actions. The same checks are also available as two scripts that say who must act on each finding:

- `Test-ExecutableSignature.ps1` : signature and certificate only. Problems the **publisher** must fix.
- `Test-ExecutableEnvironment.ps1` : everything else on this PC. Problems **you** or your **IT administrator** can fix.

Contents

- Quick start
- Parameters
- Output and exit codes
- How each check works
  - 1. File identity
  - 2. Digital signature
  - 3. Microsoft Defender scan
  - 4. App Control (WDAC, Smart App Control, AppLocker)
  - 5. File permissions
- Mark of the Web explained
- Worked example
- Runtime impact of a missing timestamp
- Fixing it at the source: jsign on a GitLab Linux runner
  - When the TSA URL is not reachable
  - Maven: signing several executables embedded in a Go installer
    - Embedded executables have no Mark of the Web

- Limitations

Quick start

```
powershell -ExecutionPolicy Bypass -File .\Test-ExecutableSafety.ps1 -Path
"$env:USERPROFILE\Downloads\setup.exe"
```

With the vendor's published hash and expected publisher:

```
.\Test-ExecutableSafety.ps1 .\setup.exe -ExpectedSha256 3A7B... -ExpectedPublisher "Contoso Ltd" -
ReportPath .\report.json
```

Works in Windows PowerShell 5.1 and PowerShell 7. It runs without Administrator rights. Running as Administrator unlocks two extra details: the Code Integrity event log and the full CiTool policy list.

Parameters

Parameter	Purpose
`-Path` (required)	The file to check (`.exe` , `.msi` , `.dll` , ...).
`-ExpectedSha256`	The SHA-256 hash the vendor publishes. A mismatch is a failure.
`-ExpectedPublisher`	Text the signer name must contain, e.g. `"Mozilla Corporation"` . A mismatch is a failure.
`-SkipDefenderScan`	Skip the Microsoft Defender scan (faster).
`-SkipRevocationCheck`	Don't contact the certificate authority to check revocation (offline use).
`-ReportPath`	Save the findings to a `.json` or `.csv` file.

Output and exit codes

Every finding gets one of four levels:

Level	Meaning
`PASS`	The check succeeded.
`INFO`	A fact worth knowing; not good or bad on its own.
`WARN`	Something to review before running.
`FAIL`	A serious problem. Don't run the file.

The verdict at the end, and the script's exit code, come from the worst level found:

Exit code	Verdict	Condition
`0`	No problems found	Only PASS and INFO findings
`1`	Review the warnings	At least one WARN, no FAIL
`2`	Do not run	At least one FAIL

The exit code makes the script usable in automation, e.g. `if ($LASTEXITCODE -eq 2) { ... }` .

How each check works

1. File identity

Hash. `Get-FileHash -Algorithm SHA256` gives the file's fingerprint. If you pass `-ExpectedSha256`, the two are compared. A match proves the file is byte-for-byte what the vendor published. The script also prints a VirusTotal lookup link built from the hash. It doesn't upload anything; you open the link yourself.

Deceptive names. Two common tricks are flagged as FAIL:

- Hidden Unicode direction characters (e.g. U+202E, "right-to-left override") that make `invoice[U+202E]fdp.exe` display as `invoiceexe.pdf`.
- Double extensions such as `report.pdf.exe`. Explorer hides known extensions by default, so this shows as `report.pdf`.

Hidden or System file attributes are a WARN.

PE header. The script reads the first bytes of the file directly with a `BinaryReader`, without loading or running it:

- `MZ` at offset 0 and `PE\0\0` at the offset stored at `0x3C` confirm it's a Windows executable.
- The *Machine* field gives the architecture (x86, x64, ARM64).
- The *Characteristics* field says whether it's a DLL.
- The *Subsystem* field says whether it's a GUI or console program.
- A non-empty CLR data directory (entry 14) means it's a .NET assembly.
- *DllCharacteristics* shows whether ASLR (`0x40`) and DEP/NX (`0x100`) are enabled. Missing ones are reported as INFO; modern compilers set both.

A PE file with an unexpected extension (e.g. a `.jpg` that is really an `.exe`) is a WARN.

Version info. `[Diagnostics.FileVersionInfo]::GetVersionInfo()` reads the CompanyName, ProductName, FileDescription, FileVersion and OriginalFilename embedded by the developer. These are self-declared, so anyone can write anything there. They're used later to cross-check against the signer.

Mark of the Web. Reads the `Zone.Identifier` stream to learn the download zone and source URL. See Mark of the Web explained. A missing mark is a WARN, because it means SmartScreen and Smart App Control won't check the file.

Extra data streams. `Get-Item -Stream *` lists all NTFS alternate data streams. Anything other than the main data, `Zone.Identifier` and `SmartScreen` is a WARN, because hidden streams can carry a payload.

2. Digital signature

Authenticode status. `Get-AuthenticodeSignature` asks Windows (`WinVerifyTrust`) whether the signature is valid:

Status	Level	Meaning
`Valid`	PASS	Signed, untampered, chains to a trusted root.
`NotSigned`	WARN	No signature, so the author can't be verified. Many open-source tools are unsigned, so this alone isn't a failure.
`HashMismatch`	FAIL	The file was changed after it was signed.
`NotTrusted` / other	FAIL	The certificate isn't trusted on this PC.

It also reports whether the signature is **embedded** (Authenticode) or comes from a **Windows catalog file** (Catalog, used for OS files), and whether the file is an OS binary.

Certificate details. The signer, subject, issuer, validity dates and thumbprint are printed. Extra checks:

- **New certificate.** A certificate issued less than 30 days ago is a WARN. Attackers sometimes buy fresh certificates for a campaign.
- **EV certificate.** If the certificate carries policy OID `2.23.140.1.3`, it's an Extended Validation code-signing certificate. The CA verified the company's identity strictly. Reported as PASS.
- **Timestamp.** A timestamp from a trusted timestamp authority (TSA) proves *when* the file was signed, so the signature stays valid after the certificate expires. No timestamp is a WARN. No timestamp *and* an expired certificate is a FAIL. See How the timestamp is detected and Runtime impact of a missing timestamp.

How the timestamp is detected

Configuring a TSA URL in the signing tool isn't enough. What counts is whether a timestamp **ended up in the file**. The signing tool sends the signature's hash to the TSA, and the TSA's signed reply is stored inside the file's signature as an *unsigned attribute*. If that request never happened or failed, there's no timestamp, whatever was configured.

The script checks in two ways:

1. **Windows' view.** `Get-AuthenticodeSignature` returns `TimeStamperCertificate`. Windows only fills this in after the signature validates, so when the chain fails (e.g. an untrusted or missing issuer) it can be empty even though a timestamp exists.
2. **Reading the file directly.** `Get-EmbeddedSignatureInfo` reads the PE *certificate table* (data directory entry 4), takes the PKCS#7 blob from the `WIN_CERTIFICATE` structure, and decodes it with .NET `SignedCms`. It then looks at the signer's unsigned attributes:

Where	OID	Meaning
Unsigned attribute	`1.3.6.1.4.1.311.3.3.1`	RFC 3161 timestamp token (`signtool /tr`). The TSA name and signing time are read from it.
Counter-signature	`1.2.840.113549.1.9.6`	Legacy Authenticode timestamp (`signtool /t`).
Unsigned attribute	`1.3.6.1.4.1.311.2.4.1`	Nested signature (dual signing, e.g. SHA-1 plus SHA-256). Counted and reported.

If either way finds a timestamp, the result is a PASS that names the TSA and the signing time in UTC. If only the direct read finds it, an INFO line says Windows didn't report it because validation failed. If the signature contains neither kind of timestamp, the WARN says the signing tool didn't add one. With signtool, that means checking that `/tr <TSA url> /td sha256` was passed and the TSA request succeeded. Other tools, such as `osslsigncode` without `-ts`, or a build pipeline that signs in a different step, can also leave it out.

The same direct read reports how many certificates are embedded in the signature. A signature that includes only the leaf certificate depends on the PC already having the issuing CA's certificate. Without it, the chain is incomplete.

Chain and revocation. The script builds the certificate chain itself with `X509Chain`:

- `RevocationMode = Online` and `RevocationFlag = EntireChain` make Windows contact each CA's CRL or OCSP server to check whether any certificate in the chain has been revoked. The timeout is 15 seconds.
- The Code Signing usage (OID `1.3.6.1.5.5.7.3.3`) is required.
- If the signature is timestamped, certificate expiry is ignored, matching how Windows treats timestamped signatures.

The extra build is needed because `Get-AuthenticodeSignature` doesn't always check revocation. A revoked certificate is the strongest warning sign there is: the CA has withdrawn it, usually because the key was stolen or used to sign malware.

Chain problem	Level
Revoked	FAIL
Untrusted root (often self-signed)	FAIL
Incomplete chain (issuer certificate not installed)	FAIL
Not valid for code signing	FAIL
Revocation server unreachable	WARN

Publisher match.

- With `-ExpectedPublisher`, the signer name or subject must contain the given text, or it's a FAIL.
- The signer is always compared with the CompanyName from version info. Both names are normalised first by stripping "Inc", "Ltd", "Corporation" etc. and punctuation. A mismatch is a WARN, because malware often copies a well-known CompanyName that the real signer doesn't match.

3. Microsoft Defender scan

- `Get-MpComputerStatus` reports whether Defender is the active antivirus or in passive mode, whether real-time protection is on, and how old the definitions are. Definitions more than 7 days old are a WARN.
- The script finds the newest `MpCmdRun.exe`, under `C:\ProgramData\Microsoft\Windows Defender\Platform\<version>\` with a fallback to `Program Files`, and runs:

  ```
  MpCmdRun.exe -Scan -ScanType 3 -File <path> -DisableRemediation
  ```

  `-ScanType 3` scans just the one file. `-DisableRemediation` reports any threat without quarantining or deleting the file.
- Exit code `0` means no threats (PASS) and `2` means a threat was found (FAIL). Anything else means the scan didn't complete (WARN).

4. App Control (WDAC, Smart App Control, AppLocker)

This section predicts whether Windows will **block** the file. Windows can't test a file against an App Control policy without running it, so this is a prediction from the policy state and the signature, not a guarantee.

App Control for Business (WDAC). Queries the `Win32_DeviceGuard` WMI class in `root\Microsoft\Windows\DeviceGuard`:

- `CodeIntegrityPolicyEnforcementStatus` covers kernel mode (drivers).
- `UsermodeCodeIntegrityPolicyEnforcementStatus` covers user mode (UMCI). **This one decides whether `.exe` files may run.**

Values: `0` Off, `1` Audit, `2` Enforced. The prediction:

UMCI	File signature	Result
Off	any	PASS: WDAC won't block it.
Audit	any	INFO: won't block, but logs event 3076 if it would have.
Enforced	not valid	WARN: block expected unless a policy allows its hash.

UMCI	File signature	Result
Enforced	valid	WARN: runs only if a policy trusts this signer, its hash or a managed installer. Ask your administrator.

On Windows 11 22H2 and later, `CiTool.exe --list-policies --json` lists the active Code Integrity policies by name. Windows' own built-in policies, such as the vulnerable driver blocklist, appear here too and only affect drivers.

Smart App Control (Windows 11 consumer feature). Read from the registry value `HKLM\SYSTEM\CurrentControlSet\Control\CI\Policy\VerifiedAndReputablePolicyState`: `0` Off, `1` On, `2` Evaluation. When it's on, unsigned or invalidly signed files are likely blocked unless Microsoft's cloud already knows them as safe. Validly signed files from a trusted CA are generally allowed.

Past block events. Searches the `Microsoft-Windows-CodeIntegrity/Operational` event log for the last 30 days for events that mention the file name:

Event ID	Meaning
3077	Blocked by an enforced App Control policy
3076	Would have been blocked (audit mode)
3033	Blocked: didn't meet the required signing level
3034	Would have been blocked: signing level (audit mode)

Reading this log needs Administrator rights.

AppLocker. Exports the effective policy with `Get-AppLockerPolicy -Effective -Xml` and checks whether any Exe, Msi or Dll rules exist. If they do, the policy is saved to a temporary XML file and the script runs:

```
Test-AppLockerPolicy -XmlPolicy <temp.xml> -Path <file> -User <you>
```

This is a real dry run: it returns `Allowed`, `Denied` or `DeniedByDefault` without executing anything. The script also reports each collection's enforcement mode and whether the Application Identity service (`AppIDSvc`) is running; AppLocker only enforces when that service runs. The AppLocker module doesn't exist on Windows Home, so there it's reported as not available.

5. File permissions

The ACL is read with `Get-Acl` and `GetAccessRules(..., [SecurityIdentifier])`. Using SIDs instead of names makes the checks work in any Windows language.

- **Owner and every access rule** are listed as INFO.
- **Can you execute it?** Your token's SIDs (your user SID plus every group from `WindowsIdentity.GetCurrent().Groups`) are matched against the rules. Allow and Deny rights are combined. A Deny on `ExecuteFile` (`0x20`) is a WARN because the file won't start. No Allow for it is also a WARN. This is an approximation; Windows' own access check also considers integrity levels and privileges.
- **Can others replace it?** If a broad group has write, delete, change-permission or take-ownership rights on the file, that's a WARN. Another user or process could swap the file between your check and your run (a "time-of-check to time-of-use" attack). The broad groups are Everyone (`S-1-1-0`), Authenticated Users (`S-1-5-11`), Users (`S-1-5-32-545`), INTERACTIVE, Guests, ANONYMOUS LOGON and NETWORK.
- **Can others plant DLLs?** The same check runs on the containing folder (create files or folders rights). Windows searches a program's own folder first when loading DLLs, so a writable folder lets someone plant a malicious DLL ("DLL hijacking").

- **Stray DLLs next to it.** If the program sits next to `.dll` files in a user folder such as Downloads, that's a WARN, because an old malicious download could be side-loaded. Move the program to its own empty folder before running it. System folders (`Windows`, `Program Files`) are skipped.

Mark of the Web explained

Mark of the Web (MotW) is the tag Windows attaches to files that come from the internet. It isn't stored in the file's contents. It's in a hidden **NTFS alternate data stream** named `Zone.Identifier`. An NTFS file can have several named streams; the normal contents live in the unnamed `:$DATA` stream, and the mark sits beside it without changing the file's size or hash.

Example contents:

```
[ZoneTransfer]
ZoneId=3
ReferrerUrl=https://example.com/downloads/
HostUrl=https://cdn.example.com/files/setup.exe
```

Field	Meaning
`ZoneId`	Security zone the file came from: `0` this computer, `1` intranet, `2` trusted sites, `3` internet, `4` restricted sites
`HostUrl`	The exact URL the file was downloaded from
`ReferrerUrl`	The page you were on when the download started

Who writes it. The program that saves the file: browsers (Edge, Chrome, Firefox), Outlook, Teams and Windows' built-in zip extraction, using the Attachment Execution Service (`IAttachmentExecute`) or by writing the stream directly. `curl`, copies from USB sticks and some archive tools don't add it.

Who reads it. SmartScreen ("Windows protected your PC"), Smart App Control, Office Protected View and macro blocking, PowerShell's `RemoteSigned` execution policy, and Explorer's *Unblock* checkbox.

Not passed on to extracted or installed files. Files a program writes, such as an installer extracting its payload, don't get the mark. See Embedded executables have no Mark of the Web.

How it's lost.

- `Unblock-File` or Properties → Unblock deletes it.
- Copying to FAT32 or exFAT drives or some network shares drops it, because those have no alternate streams.
- Some archive tools don't pass it on to files they extract. Attackers use this to bypass SmartScreen.

See it yourself:

```
Get-Item .\file.exe -Stream *                    # list all streams
Get-Content .\file.exe -Stream Zone.Identifier   # read the mark
```

In `cmd.exe`, `dir /r` shows the streams.

The mark is a clue, not proof. Anyone who can write to the file can edit or delete it.

Worked example

A vendor's client launcher downloaded from the vendor's own site produced this:

```
== Digital signature ==
  [FAIL] Signature problem (UnknownError): A certificate chain could not be built to a trusted root
authority
  [INFO] Signed by: Example Client Launcher
  [INFO] Issued by: Example Corp Internal CA UAT 1
  [WARN] Signature has no timestamp; it stops being valid when the certificate expires. The signature
block contains
         no RFC 3161 or legacy timestamp, so the signing tool did not add one (...)
  [INFO] Embedded signature: sha256 digest, 1 certificate(s) included, 0 nested signature(s).
  [INFO] Issuer certificate 'Example Corp Internal CA UAT 1' is not installed on this PC, so the chain
stops at the signer.
  [FAIL] Certificate chain is incomplete.
...
  [PASS] Defender found no threats in the file.
  [PASS] No enforced user-mode App Control policy: WDAC will not block this file.

  VERDICT: DO NOT RUN.
```

What this says:

- The file is signed, but by a company-internal CA ("Example Corp Internal CA UAT 1") that isn't trusted on this PC. "UAT" normally means a test environment. The file is probably meant for machines that have the company's root certificate installed.
- Nothing malicious was found, and nothing on this PC would block it.
- **Right action:** confirm with the vendor that this is the correct download and get the official hash or root certificate from them. Don't install an unknown root certificate yourself just to make the check pass; a root certificate is trusted for everything on the PC.

Runtime impact of a missing timestamp

While the signing certificate is valid, a missing timestamp makes no difference. It only matters once the certificate is past its expiry date (`NotAfter`), or if the certificate is revoked.

Without a timestamp, Windows checks the signature against **today's date**. After expiry it finds an expired certificate and nothing proving the file was signed while the certificate was valid. The signature then counts as **invalid** (`CERT_E_EXPIRED`, `0x800B0101`).

The file itself isn't damaged. Windows treats it like an unsigned file, so whether it still runs depends on what checks it.

What happens after expiry

Component	Effect when the signature is invalid
Normal launch (double-click, `CreateProcess`)	**Still runs.** Windows doesn't require a valid signature to start an ordinary exe.
SmartScreen (files with Mark of the Web)	Treated as an unknown publisher. Reputation earned by the certificate no longer helps, so "Windows protected your PC" is more likely. Users can still click *Run anyway* unless policy forbids it.
UAC prompt (installers and tools needing admin)	Yellow **"Publisher: Unknown"** prompt instead of the blue verified-publisher prompt. Elevation is still allowed by default.

Component	Effect when the signature is invalid
UAC "Only elevate executables that are signed and validated" (registry `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\ValidateAdminCodeSignatures = 1`)	**Blocked from elevating.** Off by default; some managed company builds turn it on.
Smart App Control	**Blocked**, unless Microsoft's cloud already rates the file as safe. It needs a valid signature or good reputation.
App Control for Business (WDAC)	**Usually still allowed.** By default, Code Integrity doesn't reject a signature because its certificate expired, so publisher rules keep matching. See the exceptions below.
AppLocker publisher rules	Generally keep matching the publisher name. Test this if your environment relies on it.
Antivirus / EDR	An invalid signature raises the file's risk score. Rules like "invalid signature running from a user folder" may alert on it or block it.
Your own install or update code	Anything that checks signatures (`WinVerifyTrust`, `Get-AuthenticodeSignature`, an updater confirming a download is genuine) **fails.** Auto-updates can stop working on the expiry date.
Explorer → Properties → Digital Signatures	"This digital signature is not valid."
`Test-ExecutableSafety.ps1`	FAIL: "Certificate has expired and there is no timestamp."

WDAC *does* reject the file when:

- The policy enables rule option **20, `Enabled:Revoked Expired As Unsigned`**. Files signed with a revoked certificate, or with an expired certificate that carries the *Lifetime Signing* EKU (`1.3.6.1.4.1.311.10.3.13`), are then treated as unsigned.
- The policy has no publisher rule for the signer at all, and only allows files by hash, by managed installer, or by Microsoft's reputation service (ISG).

A certificate with the Lifetime Signing EKU expires its signatures with the certificate **even when they are timestamped**. Check its EKUs (the script shows the certificate; `certutil -dump` or Explorer's certificate details list them). Code-signing certificates normally carry only Code Signing (`1.3.6.1.5.5.7.3.3`).

Revocation: a bigger risk than expiry

If the CA **revokes** the signing certificate, for example after a key compromise or when the certificate is retired:

	Effect
With a timestamp	Files signed *before* the revocation date stay valid. Only files signed after it become invalid.
Without a timestamp	**Every file ever signed with that certificate becomes invalid at once.** Windows can't tell which files were signed before the problem.

Smart App Control, SmartScreen and EDR react straight away, and WDAC does too if option 20 is enabled. This can happen at any time, not just on the expiry date.

Timeline for an untimestamped file

Period	What happens
Signing date until the certificate's `NotAfter`	No difference from a timestamped file, as long as the certificate chains to a trusted root.
After `NotAfter`	Smart App Control blocks it. Signature checks in install and update code fail. UAC shows "Unknown publisher". SmartScreen and EDR treat it as unsigned. Under a WDAC publisher rule, without option 20, it most likely still runs.
Any time the certificate is revoked	All of the above, immediately, for every file signed with it.

What to do

- **Before the certificate expires:** add a timestamp without re-signing, using `jsign timestamp ...` (jsign 7.0+) or `signtool timestamp /tr <TSA url> /td sha256 <file>`. Then check it with `verify-signature.sh` or `Test-ExecutableSafety.ps1`.
- **After it has expired:** a timestamp can no longer help, because it would prove signing happened after expiry. Re-sign the file with a valid certificate and a timestamp, and ship it again.
- **Check ahead of time:** look for an expiry date on untimestamped release files in your build or release checklist. To see the effect, set a test VM's clock past `NotAfter` and run `Test-ExecutableSafety.ps1`, `signtool verify /pa /v`, or the installer itself.

Fixing it at the source: jsign on a GitLab Linux runner

The worked example had two signing faults: **no timestamp** and **only the signer's certificate embedded**. Both come from how the file was signed. This section shows how to sign with jsign on a GitLab Linux runner so that neither can happen, and so the pipeline fails if they do.

Facts from the jsign documentation (version 7.6) that matter here:

Option	What to know
`--tsaurl <URL>[,<URL>...]`	Address of the timestamp authority (TSA). Several comma-separated URLs are tried in order as fallbacks. **Always set it explicitly.** The Maven plugin stopped timestamping by default in 7.6, so an upgrade can silently drop timestamps.
`--tsmode RFC3161|Authenticode`	Defaults to **Authenticode** (the older format). Set `RFC3161` explicitly; some TSAs only speak RFC 3161.
`--tsretries` / `--tsretrywait`	Defaults to 3 retries, 10 seconds apart.
`--certfile <file>`	The PKCS#7 certificate chain (PEM, or DER `.p7b` / `.spc`). Required when the key store holds only the key, as cloud key stores and HSMs often do. If it holds only the signer certificate, jsign tries to download the intermediates from the certificate's AIA URL, which fails on an isolated runner. **Supply the full chain: signer plus intermediates.**
`--proxyUrl` / `--proxyUser` / `--proxyPass`	HTTP proxy for TSA and key store traffic.
`jsign timestamp ...`	Command (jsign 7.0+) that adds a timestamp to an already-signed file without signing it again.

1. Build a signing image once

Bake jsign into an image in your own registry instead of downloading it in every job. Pin the version and check its hash; get the SHA-256 from the releases page.

```
# ci/jsign.Dockerfile
FROM eclipse-temurin:21-jre
ARG JSIGN_VERSION=7.6
ARG JSIGN_SHA256=<sha256-of-jsign-7.6.jar>

RUN apt-get update \
 && apt-get install -y --no-install-recommends curl ca-certificates openssl osslsigncode \
 && rm -rf /var/lib/apt/lists/* \
 && curl -fsSL -o /opt/jsign.jar \
      "https://github.com/ebourg/jsign/releases/download/${JSIGN_VERSION}/jsign-${JSIGN_VERSION}.jar" \
 && echo "${JSIGN_SHA256}  /opt/jsign.jar" | sha256sum -c - \
 && printf '#!/bin/sh\nexec java $JAVA_OPTS -jar /opt/jsign.jar "$@"\n' > /usr/local/bin/jsign \
 && chmod +x /usr/local/bin/jsign
```

`openssl` and `osslsigncode` are included for the reachability check and the verification step below. If your build machines trust a corporate CA (for an internal TSA, or a proxy that inspects TLS), add its root to the Java trust store here, e.g. `keytool -importcert -cacerts -noprompt -alias corp-root -file corp-root.pem`.

2. Pipeline job

```
# .gitlab-ci.yml (sign stage)
sign-exe:
  stage: sign
  image: $CI_REGISTRY_IMAGE/jsign:7.6
  needs: [build-exe]
  variables:
    # Primary TSA first, fallbacks after. Use your internal TSA if you have one.
    TSA_URLS: "http://timestamp.digicert.com,http://timestamp.sectigo.com"
    EXE: dist/app.exe
  script:
    - sh ci/check-tsa.sh "$TSA_URLS"          # fail fast if no TSA answers
    - jsign
        --storetype "$SIGN_STORETYPE" --keystore "$SIGN_KEYSTORE"
        --storepass "$SIGN_STOREPASS" --alias "$SIGN_ALIAS"
        --certfile "$SIGN_CHAIN_PEM"
        --alg SHA-256
        --tsaurl "$TSA_URLS" --tsmode RFC3161 --tsretries 5 --tsretrywait 15
        ${PROXY_URL:+--proxyUrl "$PROXY_URL"}
        "$EXE"
    - sh ci/verify-signature.sh "$EXE"        # fail if timestamp or chain is missing
  artifacts:
    paths: [dist/app.exe]
```

- `SIGN_*` are protected, masked CI/CD variables. `SIGN_CHAIN_PEM` is a **File**-type variable, so GitLab gives the job a path to a PEM file with the signer and intermediate certificates.
- `SIGN_STORETYPE` / `SIGN_KEYSTORE` depend on where the key lives, e.g. `AZUREKEYVAULT`, `PKCS11`, `DIGICERTONE`, `GOOGLECLOUD`, `AWS`. See the jsign docs for each store's values. The Windows Code Signing Playbook (in *docs/Windows-code-signing.pdf*) has an Azure Key Vault example.

- `PROXY_URL` is optional. When it's set, jsign uses it (see below).
- Sign every file in **one** `jsign` call (`jsign ... a.exe b.dll c.msi`) rather than one call per file. It's faster, and it makes fewer requests to public TSAs, which may throttle.

3. Check the TSA before signing

`ci/check-tsa.sh` sends a real RFC 3161 request to each TSA and stops the job if none of them grants it. This turns a network problem into a clear message *before* anything is signed.

```
#!/bin/sh
# Usage: check-tsa.sh "http://tsa1,http://tsa2"
set -u
ok=""
for url in $(echo "$1" | tr ',' ' '); do
  head -c 32 /dev/urandom > /tmp/tsa-probe.bin
  openssl ts -query -data /tmp/tsa-probe.bin -sha256 -cert -out /tmp/tsa-probe.tsq
  if curl -fsS --max-time 15 ${PROXY_URL:+--proxy "$PROXY_URL"} \
       -H "Content-Type: application/timestamp-query" \
       --data-binary @/tmp/tsa-probe.tsq -o /tmp/tsa-probe.tsr "$url" \
     && openssl ts -reply -in /tmp/tsa-probe.tsr -text | grep -q 'Status: Granted'; then
    echo "TSA OK:     $url"; ok=1
  else
    echo "TSA FAILED: $url"
  fi
done
[ -n "$ok" ] || { echo "No timestamp authority reachable - refusing to sign."; exit 1; }
```

The request contains only the hash of 32 random bytes, so nothing about your build is sent.

4. Verify the result before publishing

`ci/verify-signature.sh` reads the signature that ended up in the file and fails the job if there's no timestamp or the chain wasn't embedded. These are the same two checks `Test-ExecutableSafety.ps1` makes on Windows.

```
#!/bin/sh
# Usage: verify-signature.sh <file.exe> [<file2.exe> ...]
set -eu
[ "$#" -gt 0 ] || { echo "No files to verify"; exit 1; }
for f in "$@"; do
  osslsigncode extract-signature -pem -in "$f" -out /tmp/sig.pem \
    || { echo "FAIL: $f is not signed"; exit 1; }

  if openssl pkcs7 -in /tmp/sig.pem -print | grep -Eq '1\.3\.6\.1\.4\.1\.311\.3\.3\.1|countersignature';
then
    echo "OK:   $f is timestamped"
  else
    echo "FAIL: $f has no timestamp"; exit 1
  fi

  n=$(openssl pkcs7 -in /tmp/sig.pem -print_certs | grep -c '^subject=')
  if [ "$n" -ge 2 ]; then
    echo "OK:   $f has $n certificates embedded"
  else
    echo "FAIL: $f has only $n certificate embedded - pass the full chain with --certfile"; exit 1
  fi
done
```

`1.3.6.1.4.1.311.3.3.1` is the RFC 3161 timestamp, and `countersignature` is the older Authenticode timestamp. Tested against real signatures: Git for Windows (RFC 3161, 4 certificates), Chrome (Authenticode, 5 certificates), and the worked example's file (no timestamp, 1 certificate, correctly rejected).

5. Sign with a CA the users trust

A timestamp and a full chain don't help if the root CA isn't trusted on the PCs that run the file:

Who runs the file	Sign with
Only company-managed PCs	The **production** internal CA, with its root deployed to those PCs (Group Policy or Intune). Never a UAT/test CA for a production release.
Customers or external users	A certificate from a public code-signing CA (OV or EV), trusted on every Windows PC without setup.

If you use an internal TSA, the same applies to it: Windows only accepts the timestamp if the TSA's certificate chains to a root the user's PC trusts.

When the TSA URL is not reachable

Timestamping is the only step in signing that needs to reach a server outside your network. When it fails, check these, roughly in order of how often each is the cause:

Watch for	Why it happens	What to do
Outbound port 80 blocked	Most public TSAs (`http://timestamp.digicert.com`, `http://timestamp.sectigo.com`) use plain HTTP. Many runner networks only allow 443. HTTP is fine here: the TSA's reply is signed, so it can't be tampered with in transit.	Allow outbound HTTP to the TSA hosts, or use a TSA that offers HTTPS.

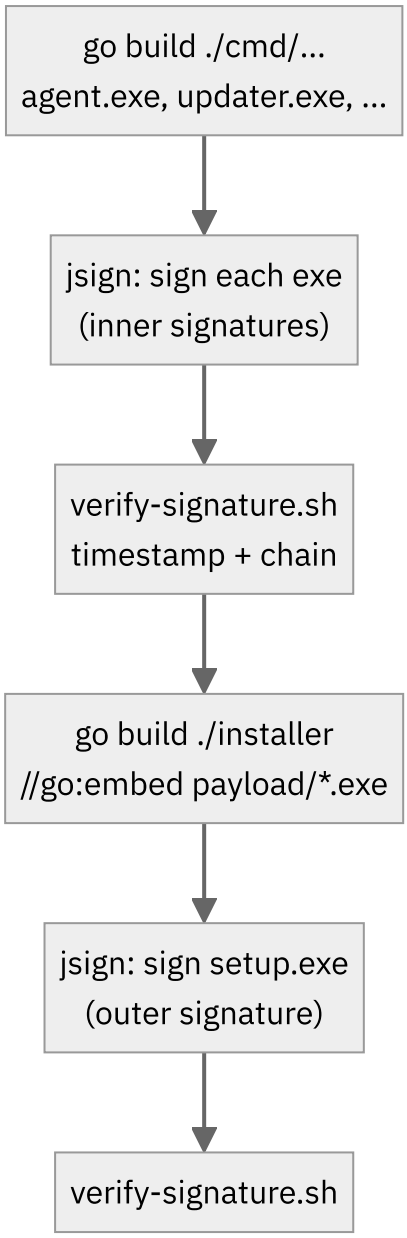
Watch for	Why it happens	What to do
Java ignores the proxy variables	GitLab runners often set `HTTP_PROXY` / `HTTPS_PROXY` / `NO_PROXY`. `curl` uses them; **Java, and so jsign, does not**. The pre-flight check can pass while jsign then fails.	Pass `--proxyUrl` (and `--proxyUser` / `--proxyPass` if needed), or set `JAVA_TOOL_OPTIONS="-Dhttp.proxyHost=... -Dhttp.proxyPort=... -Dhttps.proxyHost=... -Dhttps.proxyPort=..."`. Exclude internal hosts (e.g. an internal key store) with `-Dhttp.nonProxyHosts="*.corp.example|localhost"`, or `--nonProxyHosts` from jsign 8.0.
TLS inspection on an HTTPS TSA	A corporate proxy re-signs HTTPS traffic with its own CA, which Java's trust store doesn't know: `PKIX path building failed`.	Add the proxy's root CA to the image's Java trust store (`keytool -importcert -cacerts ...`), or use the HTTP TSA URL.
Wrong `--tsmode`	jsign defaults to Authenticode. A TSA that only speaks RFC 3161 rejects the request, which looks like a network error.	Always pass `--tsmode RFC3161`. The pre-flight check also sends RFC 3161.
DNS	Isolated runners may not resolve public names.	`getent hosts timestamp.digicert.com` in the job shows it straight away.
Throttling	Public TSAs limit bursts of requests, for example many parallel jobs each signing one file.	Sign all files in one jsign call, keep `--tsretries` / `--tsretrywait`, and list a second TSA as a fallback.
Job timeout	Each failing URL costs up to `tsretries × tsretrywait` seconds (5 × 15 s = 75 s above) before the next fallback is tried.	Size the job's `timeout:` for that, or lower the retries when you have several fallback URLs.
Internal TSA not trusted by clients	The request succeeds, but Windows on users' PCs doesn't trust the TSA's certificate, so the timestamp is ignored.	Use a public TSA for externally distributed files, or deploy the TSA's root along with your code-signing root.

Never fall back to signing without a timestamp. That file works until the signing certificate expires (usually one to three years) and then shows as invalid on every PC. If no TSA is reachable, let the job fail. If the file is already signed and only the timestamp is missing, a later job can add it without re-signing, as long as the signing certificate is still valid:

```
jsign timestamp --tsaurl "$TSA_URLS" --tsmode RFC3161 ${PROXY_URL:+--proxyUrl "$PROXY_URL"} dist/app.exe
sh ci/verify-signature.sh dist/app.exe
```

Maven: signing several executables embedded in a Go installer

This setup builds several Windows executables, signs them, embeds them in a Go installer with `go:embed`, and then signs the installer itself. That gives two layers of signatures:



- **Inner signatures** stay with each executable after the installer writes it to disk. Windows App Control, Smart App Control and Defender check them every time those programs run, long after the installer is gone.
- **The outer signature** covers the whole installer, including the embedded bytes, so it also protects the payload during download.
- **The order is fixed: inner first, outer last.** `go:embed` copies the files' bytes at build time, so signing them after `go build` changes nothing inside the installer. The installer must be signed last, because rebuilding it produces a new, unsigned file. Never modify a file after signing it (e.g. UPX compression or resource editing), because that breaks its signature.

Project layout

```
app/
├── pom.xml
├── go.mod
├── cmd/
│   ├── agent/main.go          # each becomes a Windows .exe
│   └── updater/main.go
├── installer/
│   ├── main.go                # //go:embed payload/*.exe
│   └── payload/               # build output, git-ignored
└── ci/
    ├── settings.xml
    ├── check-tsa.sh
    └── verify-signature.sh
```

`go:embed` can only embed files inside the package's own directory or its subdirectories (no `..` paths). The executables are therefore built straight into `installer/payload/`, which is listed in `.gitignore`.

Facts about the jsign Maven plugin

From the plugin source (`JsignMojo.java`) and the docs:

Item	Detail
Coordinates	`net.jsign:jsign-maven-plugin:7.6`, goal `sign`, bound to `package` by default
Files	`<file>` for one file, or `<fileset>` (a standard Maven FileSet: `directory`, `includes`, `excludes`) to sign many in one execution
Names	Same as the CLI long options, except the digest is **`<algorithm>`** (default `SHA-256`). Also `<storetype>`, `<keystore>`, `<alias>`, `<storepass>`, `<keypass>`, `<certfile>`, `<tsaurl>`, `<tsmode>`, `<tsretries>` (3), `<tsretrywait>` (10), `<name>`, `<url>`, `<replace>`, `<skip>`
Timestamp	Since 7.6 the Maven plugin **no longer timestamps by default**. Always set `<tsaurl>` and `<tsmode>RFC3161</tsmode>`; the verification step fails the build if no timestamp was added.
Passwords	`<storepass>` / `<keypass>` accept plain text, a Maven-encrypted `{...}` value, or **`mvn:<server-id>`**, which reads the password of that `<server>` in `settings.xml`
Proxy	**No proxy parameters.** The plugin uses the proxy from Maven's `settings.xml` (`<proxyId>` picks one; otherwise the first active http/https proxy), including its `nonProxyHosts`, user and password. This avoids the problem of Java ignoring `HTTP_PROXY`.

pom.xml

Plugins bound to the **same phase** run in the order they're declared in the POM, so each step gets its own phase and `jsign` is declared before `exec`:

Phase	Plugin	Step
`clean`	clean	Empty `installer/payload/`, so a stale unsigned exe can never be embedded
`initialize`	exec	`check-tsa.sh`: fail fast if no TSA answers
`compile`	exec	`go build` all `./cmd/...` into `installer/payload/`
`prepare-package`	jsign	Sign every exe in `installer/payload/`
`package`	exec	Verify the payload signatures, then `go build` the installer

Phase	Plugin	Step
`verify`	jsign, then exec	Sign `setup.exe`, then verify it

Signing is off by default (`signing.skip=true`), so developers can build without keys. The `ci-signing` profile turns it on when the `CI` variable that GitLab sets is present.

```
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>app-installer</artifactId>
  <version>1.4.0</version>
  <packaging>pom</packaging>

  <properties>
    <payload.dir>${project.basedir}/installer/payload</payload.dir>
    <dist.dir>${project.build.directory}/dist</dist.dir>
    <installer.exe>${dist.dir}/setup-${project.version}.exe</installer.exe>
    <tsa.urls>http://timestamp.digicert.com,http://timestamp.sectigo.com</tsa.urls>
    <signing.skip>true</signing.skip>
  </properties>

  <profiles>
    <profile>
      <id>ci-signing</id>
      <activation><property><name>env.CI</name></property></activation>
      <properties><signing.skip>false</signing.skip></properties>
    </profile>
  </profiles>

  <build>
    <plugins>
      <!-- mvn clean also empties the payload folder -->
      <plugin>
        <artifactId>maven-clean-plugin</artifactId>
        <configuration>
          <filesets>
            <fileset>
              <directory>${payload.dir}</directory>
              <includes><include>*.exe</include><include>*.dll</include></includes>
            </fileset>
          </filesets>
        </configuration>
      </plugin>

      <!-- Declared BEFORE exec so that, in the verify phase, signing runs before checking -->
      <plugin>
        <groupId>net.jsign</groupId>
        <artifactId>jsign-maven-plugin</artifactId>
        <version>7.6</version>
        <configuration>
          <skip>${signing.skip}</skip>
          <storetype>${env.SIGN_STORETYPE}</storetype>
          <keystore>${env.SIGN_KEYSTORE}</keystore>
          <alias>${env.SIGN_ALIAS}</alias>
          <storepass>mvn:code-signing</storepass>       <!-- server id in settings.xml -->
          <certfile>${env.SIGN_CHAIN_PEM}</certfile>    <!-- signer + intermediates -->
          <algorithm>SHA-256</algorithm>
          <tsaurl>${tsa.urls}</tsaurl>
```

```
          <tsmode>RFC3161</tsmode>
          <tsretries>5</tsretries>
          <tsretrywait>15</tsretrywait>
          <name>Example App</name>
          <url>https://www.example.com</url>
        </configuration>
        <executions>
          <execution>
            <id>sign-payload</id>
            <phase>prepare-package</phase>
            <goals><goal>sign</goal></goals>
            <configuration>
              <fileset>
                <directory>${payload.dir}</directory>
                <includes><include>*.exe</include><include>*.dll</include></includes>
              </fileset>
            </configuration>
          </execution>
          <execution>
            <id>sign-installer</id>
            <phase>verify</phase>
            <goals><goal>sign</goal></goals>
            <configuration><file>${installer.exe}</file></configuration>
          </execution>
        </executions>
      </plugin>

      <plugin>
        <groupId>org.codehaus.mojo</groupId>
        <artifactId>exec-maven-plugin</artifactId>
        <version>3.5.0</version>
        <configuration>
          <environmentVariables>
            <GOOS>windows</GOOS>
            <GOARCH>amd64</GOARCH>
            <CGO_ENABLED>0</CGO_ENABLED>
          </environmentVariables>
        </configuration>
        <executions>
          <execution>
            <id>check-tsa</id>
            <phase>initialize</phase>
            <goals><goal>exec</goal></goals>
            <configuration>
              <skip>${signing.skip}</skip>
              <executable>sh</executable>
              <arguments><argument>ci/check-tsa.sh</argument><argument>${tsa.urls}</argument>
</arguments>
            </configuration>
          </execution>

          <!-- -o with a trailing slash writes one exe per main package into the folder -->
          <execution>
```

```
            <id>build-payload</id>
            <phase>compile</phase>
            <goals><goal>exec</goal></goals>
            <configuration>
              <executable>go</executable>
              <arguments>
                <argument>build</argument>
                <argument>-trimpath</argument>
                <argument>-ldflags=-s -w -X main.version=${project.version}</argument>
                <argument>-o</argument>
                <argument>${payload.dir}/</argument>
                <argument>./cmd/...</argument>
              </arguments>
            </configuration>
          </execution>

          <!-- Same plugin, same phase: runs in declaration order (verify, then build) -->
          <execution>
            <id>verify-payload</id>
            <phase>package</phase>
            <goals><goal>exec</goal></goals>
            <configuration>
              <skip>${signing.skip}</skip>
              <executable>sh</executable>
              <arguments>
                <argument>-c</argument>
                <argument>sh ci/verify-signature.sh "${payload.dir}"/*.exe</argument>
              </arguments>
            </configuration>
          </execution>
          <execution>
            <id>build-installer</id>
            <phase>package</phase>
            <goals><goal>exec</goal></goals>
            <configuration>
              <executable>go</executable>
              <arguments>
                <argument>build</argument>
                <argument>-trimpath</argument>
                <argument>-ldflags=-s -w -X main.version=${project.version}</argument>
                <argument>-o</argument>
                <argument>${installer.exe}</argument>
                <argument>./installer</argument>
              </arguments>
            </configuration>
          </execution>

          <execution>
            <id>verify-installer</id>
            <phase>verify</phase>
            <goals><goal>exec</goal></goals>
            <configuration>
              <skip>${signing.skip}</skip>
```

```
              <executable>sh</executable>
              <arguments><argument>ci/verify-signature.sh</argument><argument>${installer.exe}
</argument></arguments>
            </configuration>
          </execution>
        </executions>
      </plugin>
    </plugins>
  </build>
</project>
```

Notes:

- If `verify-payload` fails (no timestamp, missing chain), the build stops **before** `go build` embeds anything, so a badly signed payload can never end up in an installer.
- `-ldflags=-s -w` strips debug data, which is done before signing and is safe. Don't post-process the files after jsign.
- For an ARM64 build too, add a second build execution with `GOARCH=arm64` into a separate payload folder.

installer/main.go (embedding)

```
package main

import (
    "embed"
    "io/fs"
    "os"
    "path/filepath"
)

// The exes were signed before this package was compiled; their bytes,
// signatures included, are copied into the installer unchanged.
//
//go:embed payload/*.exe
var payload embed.FS

func install(dest string) error {
    if err := os.MkdirAll(dest, 0o755); err != nil {
        return err
    }
    return fs.WalkDir(payload, "payload", func(path string, d fs.DirEntry, err error) error {
        if err != nil || d.IsDir() {
            return err
        }
        data, err := payload.ReadFile(path)
        if err != nil {
            return err
        }
        return os.WriteFile(filepath.Join(dest, d.Name()), data, 0o755)
    })
}
```

Install into a folder only administrators can write to, such as `C:\Program Files\<Product>`, not `%TEMP%` or Downloads. Otherwise another user or process could replace an extracted exe or plant a DLL next to it before it runs. This is the same risk `Test-ExecutableSafety.ps1` checks for.

Extracted files carry no Mark of the Web, so SmartScreen won't check them again. App Control and Smart App Control still check their signatures, which is why the inner signatures matter. See Embedded executables have no Mark of the Web.

Embedded executables have no Mark of the Web

The downloaded installer has a Mark of the Web; the programs it extracts don't. Go's `os.WriteFile` doesn't copy the `Zone.Identifier` stream, and Windows doesn't pass it on to files a program creates. Tested: a `setup.exe` with `ZoneId=3` extracted `agent.exe` and `updater.exe` that had only their normal `:$DATA` stream.

This is normal. Windows Installer (MSI) and other installers behave the same way. The mark protects the *download*. Once the user has accepted the installer, trust passes from the installer to what it installs.

Check	Downloaded installer (has the mark)	Embedded exes after extraction (no mark)
SmartScreen reputation check ("Windows protected your PC")	Checked when launched from Explorer	**Never checked**
"Open File – Security Warning" prompt	Can appear	Never appears
Smart App Control	Checked	**Still checked** on every launch; it doesn't depend on the mark
WDAC / App Control for Business	Checked	**Still checked**; it doesn't depend on the mark
AppLocker	Checked	Still checked
Defender / EDR	Scanned on write and on launch	Scanned on write and on launch, like normal installed software
UAC prompt (if it needs admin)	Shows the publisher from the signature	Same; it depends only on the signature
`Test-ExecutableSafety.ps1`	Shows the source URL	WARN "No Mark of the Web", which is expected here

What this means:

1. **The installer is the only file SmartScreen checks.** Its outer signature and reputation carry the whole download. A file signed by an EV certificate, or one with an established signing reputation, gets through SmartScreen most easily.
2. **The inner signatures do the work afterwards.** From the moment of extraction, only signature-based checks protect the embedded exes: Smart App Control, WDAC, EDR, and your updater's own signature check. That's why each embedded exe must be signed **and timestamped**. After the certificate expires, a payload exe without a timestamp is still not re-checked by SmartScreen, but **is blocked by Smart App Control**, fails signature checks in your updater, and looks suspicious to EDR. See Runtime impact of a missing timestamp.
3. **Where you extract matters more.** Nothing will warn about these files later. If they land in a folder other users can write to (`%TEMP%`, Downloads), someone could replace them or plant a DLL, and no prompt would catch it. Extract to Program Files.
4. **WDAC managed installer.** In a managed company environment, IT can designate the installer as a *managed installer*. Every file it writes is then tagged (in an NTFS extended attribute) and allowed by WDAC automatically. That's the WDAC way of passing on trust, instead of the mark.

5. **Don't add the mark to installed files yourself.** Every launch of your installed programs would then trigger SmartScreen or the security warning. Users would learn to click through warnings, and your own updater could break.
6. **Scripts are a special case.** If the payload contained `.ps1`, `.js` or Office files, they would also arrive without the mark. PowerShell's `RemoteSigned` wouldn't block an unsigned script, and Office wouldn't open a document in Protected View. Sign PowerShell scripts too, or keep the payload to signed exes and DLLs.

ci/settings.xml

```
<settings>
  <servers>
    <!-- read by <storepass>mvn:code-signing</storepass>; value comes from a masked CI variable -->
    <server>
      <id>code-signing</id>
      <password>${env.SIGN_STOREPASS}</password>
    </server>
  </servers>

  <!-- Only if the runner must use a proxy. jsign takes TSA and key-store traffic through it. -->
  <proxies>
    <proxy>
      <id>corp</id>
      <active>true</active>
      <protocol>http</protocol>
      <host>proxy.corp.example</host>
      <port>8080</port>
      <nonProxyHosts>*.corp.example|localhost</nonProxyHosts>
    </proxy>
  </proxies>
</settings>
```

GitLab job

The image needs Maven, a JDK, Go, `openssl` and `osslsigncode`:

```
# ci/maven-go-sign.Dockerfile
FROM maven:3.9-eclipse-temurin-21
# Pin to the Go version in your go.mod
COPY --from=golang:1.26 /usr/local/go /usr/local/go
ENV PATH="/usr/local/go/bin:${PATH}"
RUN apt-get update \
 && apt-get install -y --no-install-recommends openssl osslsigncode curl \
 && rm -rf /var/lib/apt/lists/*
```

```
build-installer:
  stage: build
  image: $CI_REGISTRY_IMAGE/maven-go-sign:latest
  variables:
    MAVEN_OPTS: "-Dmaven.repo.local=$CI_PROJECT_DIR/.m2/repository"
  cache:
    paths: [.m2/repository]
  script:
    - mvn -B -s ci/settings.xml clean verify
  artifacts:
    paths: [target/dist/*.exe]
  rules:
    - if: $CI_COMMIT_REF_PROTECTED == "true"   # signing secrets are on protected branches/tags only
```

`SIGN_STORETYPE`, `SIGN_KEYSTORE`, `SIGN_ALIAS` and `SIGN_STOREPASS` are protected, masked CI/CD variables. `SIGN_CHAIN_PEM` is a File-type variable. The `ci-signing` profile switches on automatically because GitLab sets `CI=true`.

Check a release on Windows with `Test-ExecutableSafety.ps1` on `setup-<version>.exe`. To check the embedded programs, run the installer into a test folder and run the script on each extracted exe. A "No Mark of the Web" warning on those is expected.

Limitations

- **No check proves a file is safe.** A clean result means none of these checks found a problem. Signed malware exists, and so does malware Defender doesn't detect yet.
- **App Control results are predictions.** Windows has no dry-run API for WDAC or Smart App Control. Only AppLocker can be truly tested.
- **SmartScreen reputation can't be queried** from a script. It's only checked when the file is launched.
- **Only the primary signature is examined.** Files with several signatures (e.g. SHA-1 plus SHA-256) are judged by the first.
- **The execute-permission check is an approximation** based on the ACL and your groups.
- **Revocation checks need internet access.** Without it they're a WARN, or use `-SkipRevocationCheck`.
- Some details (the Code Integrity log, the full CiTool output) need Administrator rights.